

# KLM SCROD SRAM Expansion

## Feb. 2026 Update



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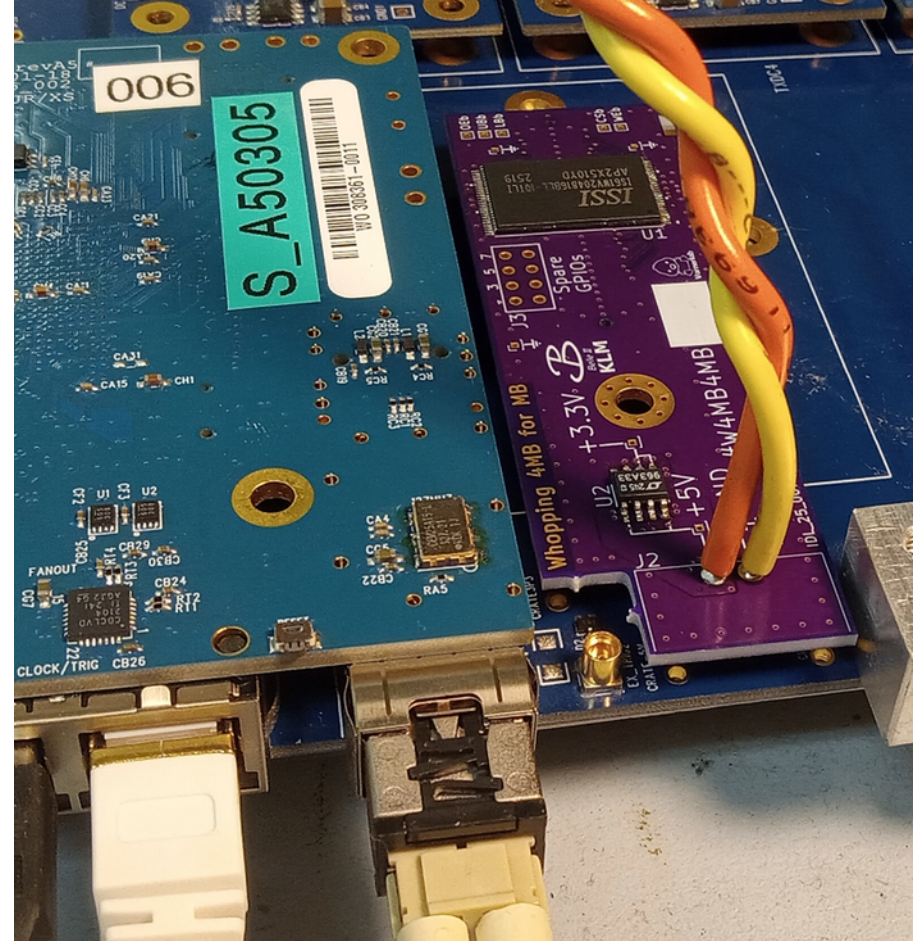


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- ☕ In events with multi-strip hits, SRAM access for fetching stored pedestals can become the bottleneck in waveform processing
- ☕ Scintillator motherboard has spare 80 pin connector with nets connected to FPGA GPIOs
- ☕ Got approval to create SRAM expansion PCB
  - ☕ Now each axis (called Bus A / Bus B on motherboard) will have its own SRAM chip for pedestal storage
  - ☕ New scheme will halve the time for pedestal access

# SRAM UPDATE

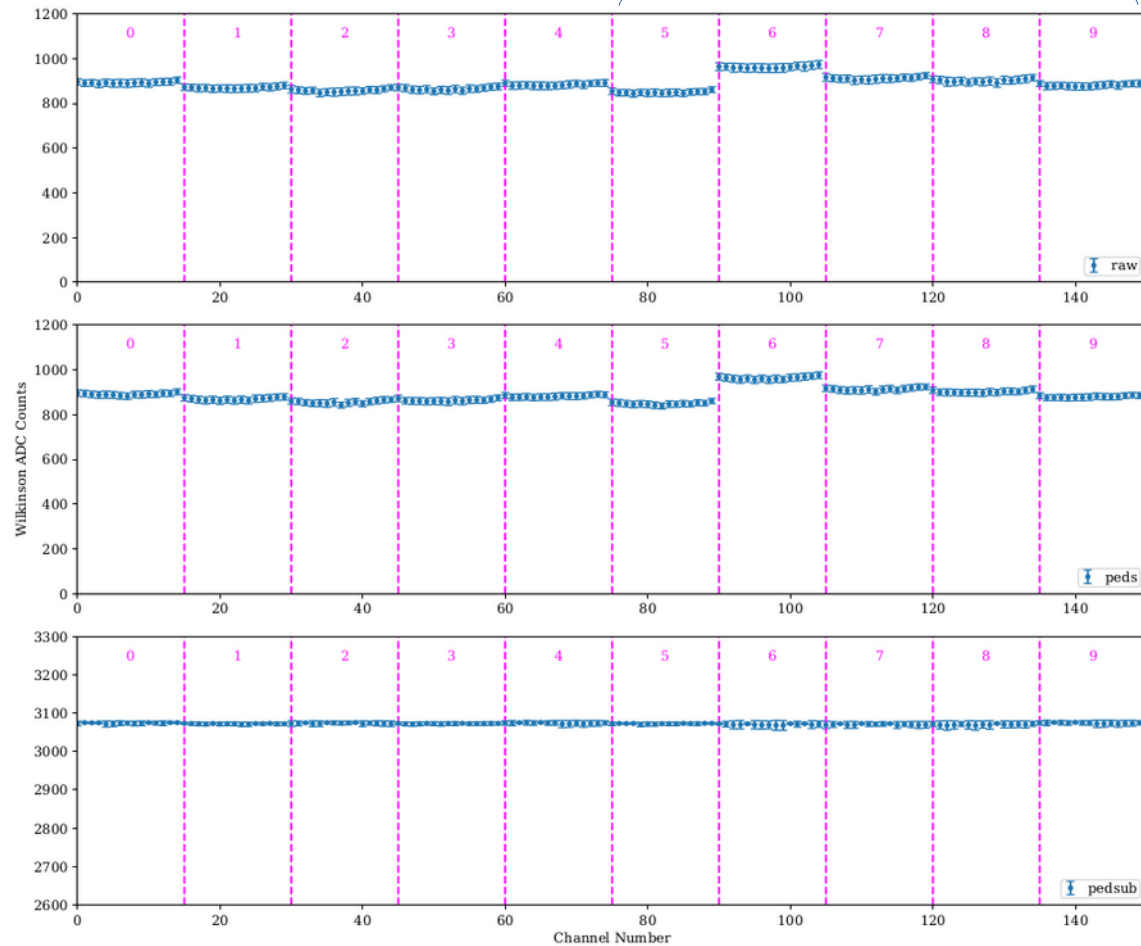
- ☕ Prototype board designed
- ☕ 1st article soldered together
  - ☕ Still finalizing choice for 5V connector
- ☕ New firmware prepared last summer by Horacio and I
- ☕ After modifying the pin mapping in firmware, our modified SRAM access logic in the SCROD firmware worked great



# TEST RESULTS (128 KSAMPLES)

Results are consistent with similar scan using only the original SRAM chip for pedestal storage!

Bus B (New SRAM)



# VERSION LOG



v3201



First compile of SRAM upgrade firmware



v3202



Tied **BUS\_REGCLR** to GND as in master branch



v3203



Fix pin mapping in UCF file



v3301

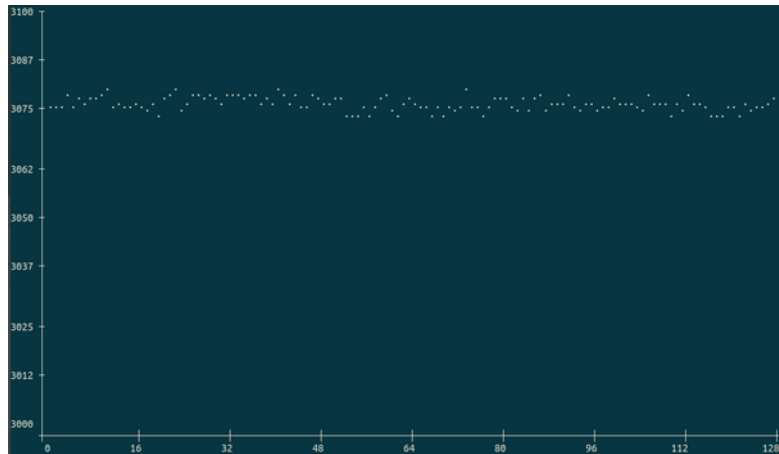


Merge with vFE33 and  
**feature/ramp\_settle**

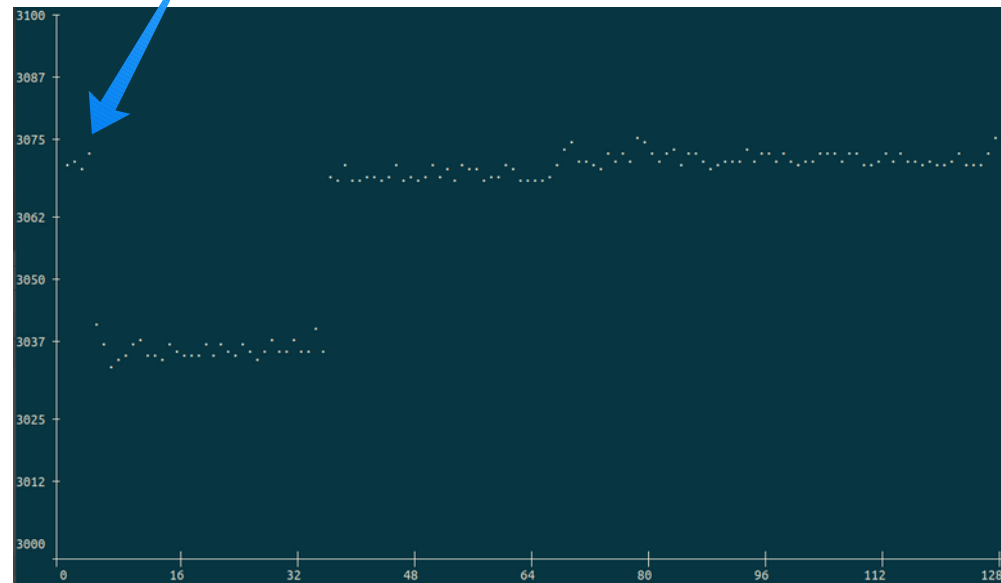
What's that?

# ramp\_settle EXPLAINED

Normal



First window only  
shifts last 4 samples



Ramp reactivates too soon --- higher starting voltage ==> fewer ADC counts

# VERSION LOG



v3201



First compile of SRAM upgrade firmware



v3202



Tied **BUS\_REGCLR** to GND as in master branch



v3203



Fix pin mapping in UCF file



v3301



Merge with vFE33 and **feature/ramp\_settle**

- **ramp\_settle** feature not working as anticipated (Wilkinson counter starts too early)



v3401



Merge with vFE34



v3402



Refactor **feature/ramp\_settle**  
(delay **BUS\_CLR** and **BUS\_RD\_ENA** along with **BUS\_RAMP**)

Now 2nd-window ADC performance is restored!

... but

Latest bit file shows outliers in pedestals from new SRAM (hardware seems okay --- need to investigate further)

# REMAINING TASKS



## SRAM Expansion PCB

- Finalize +5V connector choice  
(Cheap, easy, secure, non-interference)
- Double-check PCB design  
(Characteristic impedance? Image-current path? Happy with silkscreen?)
- Order 3 more, solder & test, then order production run



## Design RTM board for +5V supply



## Design wire harness

(Determine gauge from power budget, appropriate flexibility or stiffness, how to sheath and preform bends)



## Investigate feasibility of securing SRAM PCB with standoff

- Enough threads for proper engagement?
- Necessary screw sizes available to fully utilize threads?
- Screw heads don't interfere with neighbor boards?



## Investigate SCROD fastening hardware at same time

- Same considerations as above
- Ease of installation?



## Order all hardware, wire spools, and ICs



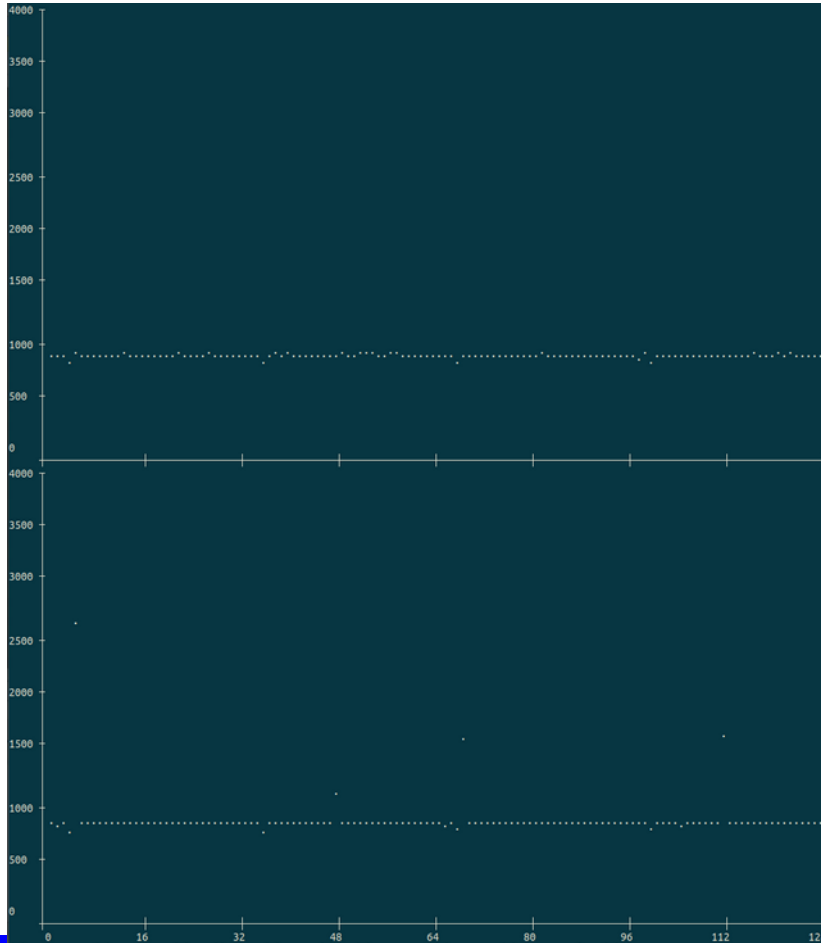
## Design clamp tool for easy MB removal

# TIMELINE

| Jan                   | Feb     | Mar                          | Apr                                                    | May | Jun | Jul                    | Aug | Sep | Oct | Nov | Dec |
|-----------------------|---------|------------------------------|--------------------------------------------------------|-----|-----|------------------------|-----|-----|-----|-----|-----|
| Soldered<br>1st board |         |                              |                                                        |     |     |                        |     |     |     |     |     |
|                       | Testing |                              |                                                        |     |     |                        |     |     |     |     |     |
|                       |         | Finalize<br>board<br>designs |                                                        |     |     |                        |     |     |     |     |     |
|                       |         |                              | -----Production run-----<br>(Send to board fab. house) |     |     |                        |     |     |     |     |     |
|                       |         |                              | Build Wiring Harnesses in house<br>(12 required)       |     |     | -----Installation----- |     |     |     |     |     |
|                       |         |                              |                                                        |     |     |                        |     |     |     |     |     |

Backup

# PEDESTAL OUTLIERS



- ☕ Top: Bus A  
Bottom: Bus B (new SRAM)
- ☕ Streaming pedestal values only (direct from SRAM chips to debug wave FIFO)
- ☕ Four outliers in this example waveform (typical)
- ☕ Reverting to previous bitfile fixed issue, so new hardware is not the issue
- ☕ Firmware logic changes completely unrelated
- ☕ Suspect cross-talk from unconstrained GPIO pins --- will investigate on oscilloscope